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Dynamic balance of heat and mass in high power density laser welding

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Abstract: The experimental system to observe the inner phenomenon of keyhole and molten pool through glass plate by high speed camera and spectrometer in high power density laser welding was set up. Two circular flows in the molten pool were observed by high speed camera, which transferred the mass from the front to the rear of the keyhole to keep the mass balance. Temperature distribution in the molten pool was firstly detected by spectrometer, which indicated that the circular flows acted as the cooling system to take heat away from the keyhole. Porosity formation process was also observed and the mechanism was discussed.

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OCIS codes: (140.3390) Laser materials processing; (140.0140) Lasers and laser optics; (300.6170) Spectra; (300.6360) Spectroscopy, laser; (150.0155) Machine vision optics.

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1. Introduction

Laser manufacturing has been widely used in the field of automobile, ship, aircraft, train, and so on, because of the high efficiency and quality [1-7]. For the high power density situation (e.g. $>1.0 \times 10^6$ W/cm² for steel), metal irradiated by laser is evaporated, then keyhole full of metal vapor penetrating into the metal is formed [8-14]. Keyhole acts as an effective tool to absorb the laser energy because of the multiple reflections of light on the wall of keyhole [11,15,16], and heat is transferred from keyhole to molten pool to offset the dissipated heat, keeping the heat balance. The keyhole and molten pool move with laser beam, metal flows appear in the molten pool to fill the previous position of the keyhole, keeping the mass balance. This dynamic balance of heat and mass plays a significant role for keeping high quality in laser manufacturing. However, the molten pool and keyhole are surrounded by the solid metal, and it is difficult to obtain the inner phenomenon by conventional experimental method. Thus, the routes of metal flows to fill the keyhole and heat dissipation away from the keyhole have rarely been described.

Various simulation researches have focused on dynamics of keyhole [17], flow of molten pool [18], and temperature distribution [17-20] in laser penetration welding. However, the physical process includes multiple reflections of light in the keyhole, evaporation of the liquid metal, re-solidification of the metal vapor, flow of the liquid metal, and interaction among metal vapor, liquid metal and air [8]. It is a great challenge to have a comprehensive

consideration of all these phenomena at present. Thus, the mechanism of heat and mass balance in laser penetration welding is not totally revealed by simulation at current stage.

The dynamic keyhole profile was observed based on a butt-joint configuration assembled from transparent glass and specimen [10, 21-23]. The results in these literatures showed that it was possible to observe the flow routes in the molten pool with a specially designed structure. X-ray transmission method has been used to trace the movement of metal in the molten pool during laser penetration welding [24], and circular flows have been detected [25, 26]. However, effects of circular flows on heat and mass balance during the welding process haven't been described.

In this paper, a specially designed experimental system was set up to observe the inside phenomena through the glass plate during the laser welding process, dynamics of keyhole and flows in the molten pool were observed by high speed camera, and temperature in the molten pool was measured by spectrometer [27, 28]. Based on the experimental results output by the newly developed experimental system, the heat and mass transfer mechanism in laser penetration welding was firstly determined. And the possible reason for porosity formation was discussed. This paper provided experimental insights into the dynamic balance of heat and mass in laser penetration welding.

2. Experimental procedures

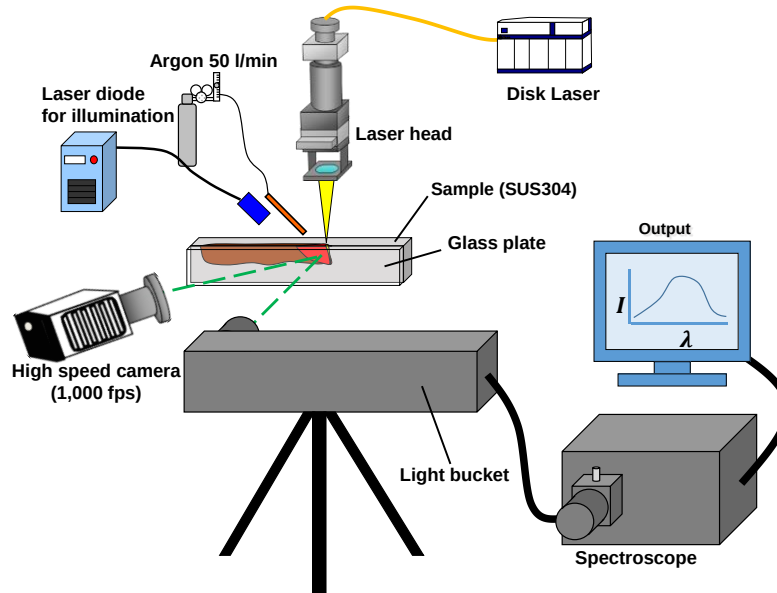


Fig. 1. Schematic illustration of experimental system for observation and spectroscopic measurement of molten pool through glass plate during laser welding.

Figure 1 shows the schematic illustration of the developed experimental system for observation and spectroscopic measurement of inside phenomenon through glass plate during laser welding. The SUS 304 steel plate was laminated with the 2 mm thick borosilicate glass plate, and high power density laser was set to irradiate at the interface between steel and glass plate. As the glass plate wouldn't melt during the experiment, this provided an effective method to observe the dynamics of keyhole and molten pool. A high speed camera of 1000 fps was used to observe the dynamics of the molten pool. A diode laser with the wavelength of 976 nm and power of 30 W was used for illumination, ensuring obtaining clear video of keyhole and molten pool by the high speed camera. A disk laser device with a maximum power of 16 kW was used in the experiment, and beam parameter product (BPP) was 8.0 mm*mrd. Diameter of the laser beam at the focus position was 280 μm. During the

experiment, the laser power was set at 6 kW, the welding speed was set at 3 m/min, and the laser beam was focused on the upper surface of the sample. The argon shielding gas was fed at a rate of 50 l/min. The SUS 304 steel with the dimension of $30 \times 9 \times 3$ mm was used in the experiment. The spectrometer PMA-50 (Hamamatsu Photonics CO. Ltd., Hamamatsu, Japan) was used to measure the temperature, which was determined by fitting the thermal radiation spectrum between 250 and 830 nm to the Planck radiation function [27, 28]:

$$I_{\lambda} = c_1 \varepsilon(\lambda) \lambda^{-5} / \left[\exp\left(\frac{c_2}{\lambda T}\right) - 1 \right] \quad (1)$$

I_{λ} was the spectrum intensity, ε was the emissivity with the value of 1.0 for black body, λ was the wavelength, $c_1 = 3.7418 \times 10^{-16} \text{ Wm}^2$, $c_2 = 0.014388 \text{ mK}$. Then the normalized spectrum intensity would be compared with that of the measured. During the experiment, the spectrometer was focused on the target point. Thus, temperature of the target point during the welding process could be obtained, and the average temperature from 600 ms to 900 ms was used to represent the temperature of the target point in the molten pool during the welding process. As the spectrometer can only focus on a target point at a time, several repetitive welding experiments with the spectrometer focused on different target points were conducted to measure the temperature distribution in the molten pool.

3. Results and discussion

3.1 Dynamic characteristics of keyhole and molten pool

The irradiation position of laser beam was presented at the top of Fig. 2. At the initial position, the laser beam focused at the interface between metal and glass plate. At the final position, the laser beam focused at the metal side. And the laser beam moved from the initial position to the final position. By changing the focus position of laser beam, dynamic characteristics of keyhole and molten pool at different longitudinal sections could be observed through the glass plate.

At 0 ms, laser started to irradiate on the surface of the material, as shown in Fig. 2(a). Metal temperature increased rapidly when absorbing the laser energy, and this part of metal was evaporated instantaneously, which led to the formation of keyhole, as shown in Fig. 2(b). The shape of the keyhole was observed to change violently. On the front wall of the keyhole, laser irradiated on the melt metal, part of the metal was evaporated immediately, while the residual liquid metal got a high flowing-down speed under the function of recoil pressure, this phenomenon was also reported by Matsunawa [12]. When the liquid metal flowed down, the volume of the bottom of the keyhole decreased, metal vapor at the bottom of the keyhole was compressed, then this compressed metal vapor obtained a high reversed upward speed. Wavy surface appeared at the rear wall of the keyhole when high speed metal vapor flowed on the surface of liquid metal. Then the heat was transferred from keyhole to the metal gradually, and metal around the keyhole was melted, as shown in Figs. 2(c)-2(e). During the experiment, center of the laser moved from the interface between the steel and glass to the steel side, and the observed size of the keyhole decreased because only a small section of the keyhole was observed, as shown in Figs. 2(f)-2(h).

Flows in the molten pool could be clearly observed through the glass plate (as shown in Visualization 1). The flow routes in the molten pool were presented in Figs. 2(f)-2(h). Two circular flows in the molten pool were firstly observed by high speed camera in our experiment. One flow started from the bottom of the keyhole and flowed along the clockwise direction. This flow was generated because the liquid metal in the keyhole had this flow direction, and this part of liquid metal flowed from keyhole into the molten pool. The other flow started from the outlet of the keyhole, and flowed along the anticlockwise direction.

Besides, some liquid metal was taken out of the keyhole by the metal vapor, and the spatter formed, as shown in Figs. 2(c)-2(h).

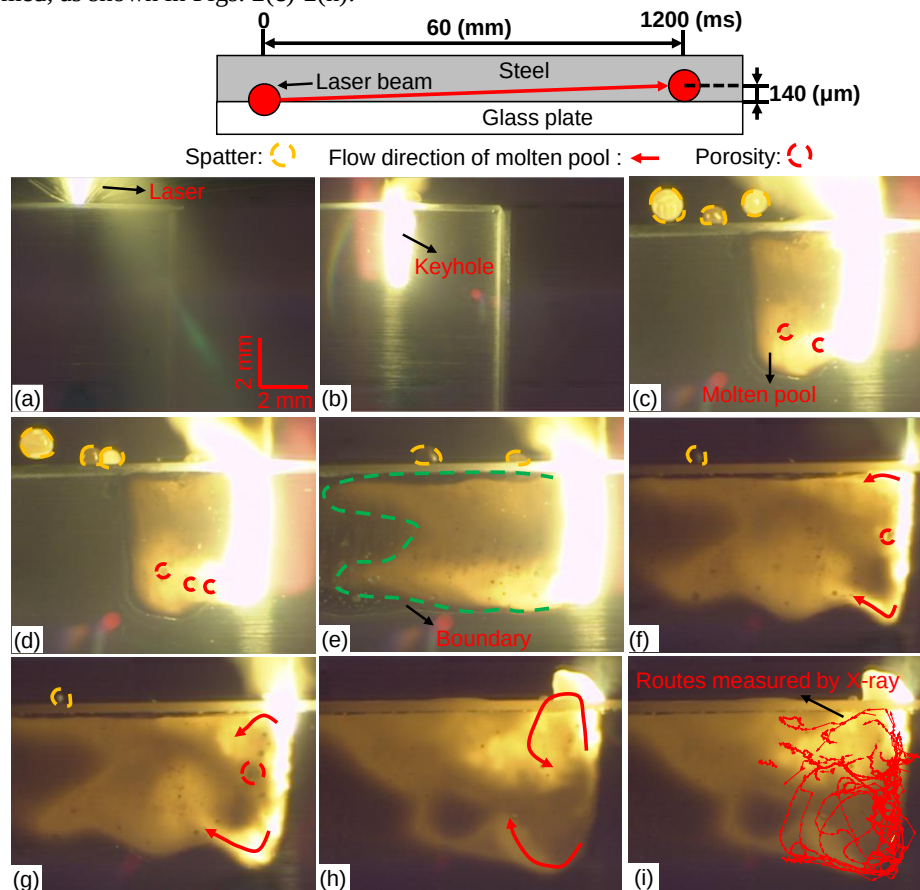


Fig. 2. Observed dynamics of keyhole and molten pool with the time and laser center at: (a) 0 ms, 0 μm; (b) 30 ms, 3.5 μm; (c) 102 ms, 11.9 μm; (d) 112 ms, 13.1 μm; (e) 400 ms, 46.7 μm; (f) 673 ms, 78.5 μm; (g) 689 ms, 80.4 μm; (h) 900 ms, 105 μm; (i) 900 ms, 105 μm, combined with the flow routes recorded by X-ray transmission method [25] (see Visualization 1).

To some extent, the designed specimen structures made of laminated steel and glass plate in this manuscript was different from the normal welding situation. The glass plate had a high reflectivity for laser, and the thermal conductivity of glass was lower than that of steel, which would affect the dynamic characteristics and temperature distribution during the welding process. To validate the availability of this developed method, the observed flows of the molten pool were compared with those measured by the three-dimensional X-ray transmission method in normal welding situation [25]. To make the experimental results from two measurement methods comparable, the same laser parameters (laser power, welding speed, defocus distance, and so on) were adopted in the experiments. As shown in Fig. 2(i), routes of the metal in the molten pool recorded by the tracers (red line) [25] were put on the captured picture of the molten pool at 240 ms. Two circular flows in the molten pool were also detected by the X-ray transmission method, similar with the flow routes observed by the newly developed method, proving that the experimental results using this newly developed setup were verified.

Then the complete mass balance process in high power laser welding of steel could be described as Fig. 3. The keyhole moved forward with the laser beam, and two circular flows transferred the melt from the front to the rear of the keyhole, which maintained the mass

balance. The liquid metal flowed down from the front wall of the keyhole, which induced the circular flow at the bottom of the molten pool. As the outlet zone had the lowest temperature in the keyhole, metal vapor re-solidified and attached at this zone. As this part of the reattached metal had the velocity with the upward and leftward direction, the anticlockwise flow was produced, which induced the circular flow at upside of the molten pool. The spatters were produced at the rear wall of the keyhole and dropped into the molten pool from the upper surface, which acted as an additional part of the mass balance system during laser welding.

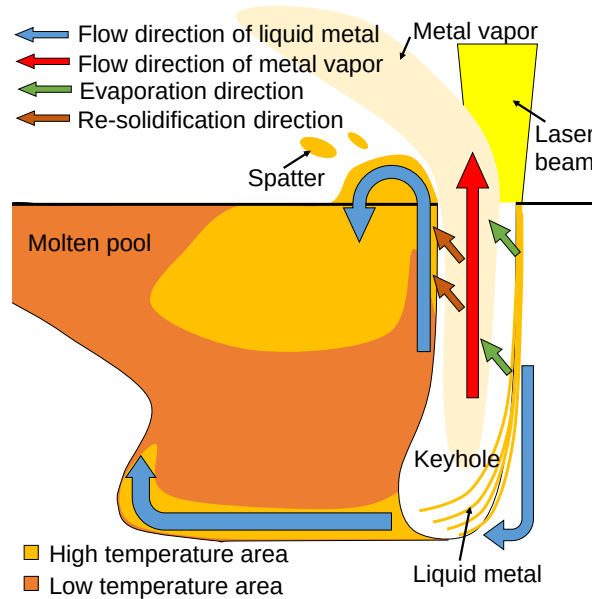


Fig. 3. Schematic of mass balance in high power density laser welding.

3.2 Molten pool temperature

Though a lot of simulation results about temperature distribution in the molten pool have been presented [19, 20], this hasn't been validated by direct experiment. In this section, the spectrometer [27] was used to measure the temperature in the molten pool through the glass plate, as shown in Fig. 4, which was the first report on temperature distribution in the molten pool of laser welding from experiment. The measured spectrum intensity was compared with that of the black body at a certain temperature, e.g. 1900, 2000, and 2100 K. As shown in Fig. 4(a), this measured spectrum intensity fit best with the emission intensity of black body at the temperature of 2000 K, then temperature value of 2000 K could be determined. The measured temperature in the molten pool was marked in Fig. 4(b). The experimental results showed that higher temperature appeared at the routes of the circular flows, which validated that the circular flows in the molten pool took the heat in the keyhole into the molten pool, keeping the dynamic heat balance. Thus, the circular flows acted as the cooling system for laser welding process.

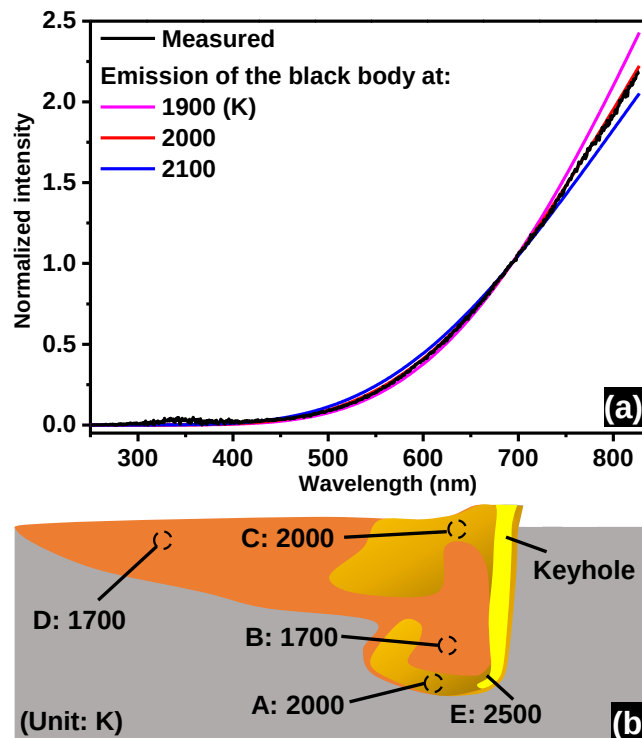


Fig. 4. (a) Comparison of the spectrum values; (b) measured temperature in molten pool.

3.3 Mechanism of porosity formation

Porosity formation in the laser welding process has been widely concentrated because it would reduce the integrality of the joint [29-32]. The X-ray computed tomography was used to measure the porosity distribution in the laser welded specimen, and simulation model considering heat transfer and fluid flow was developed to analyze the porosity formation process [29]. The simulation model indicated that the unstable flow of the molten pool led to the porosity formation. However, the originality of porosity has rarely been observed by experiment. In this section, the result of porosity initiating at the interface between keyhole and molten pool observed by high speed camera was presented, and the mechanism of porosity formation was discussed.

A part of the porosity was observed to initiate in the flow starting from the tip of the keyhole, as shown in Figs. 2(c) and 2(d). The liquid metal in front of the keyhole flowed down, and into the molten pool from the tip of the keyhole. Some air was involved into the flow because of the fluctuation in flow speed and volume of fluid, and the flow impact which included surface tension, gravity, inertia and viscous damping [33]. The other part of the porosity was observed to initiate from the rear wall of the keyhole, as shown in Figs. 2(f) and 2(g). Schematic of porosity formation process in this situation was shown in Fig. 5. Metal vapor in the keyhole flowed at a high speed along the rear wall of the keyhole, and wavy surface appeared, as shown in Fig. 5(a). In some cases, shape of the keyhole changed violently, and the wavy surface penetrated deeply into the molten pool, as shown in Fig. 5(b). When the violently wavy surface disappeared, a part of air might remain in the molten pool, as shown in Fig. 5(c). Then the porosity would form at the air zone if this air couldn't escape away from the molten pool or dissolve in the molten pool before solidification.

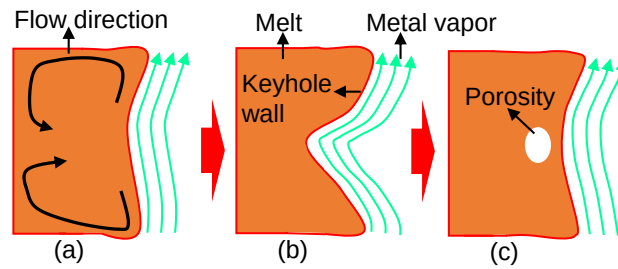


Fig. 5. Schematic of porosity formation process.

4. Summary and conclusions

Heat and mass balance system in laser welding was firstly reported based on the measured inner phenomena of keyhole and molten pool. Two circular flows were captured by high speed camera through glass plate. These circular flows transferred the mass from the front to the rear of the keyhole, keeping the mass balance in laser welding for high quality. The measured temperature results showed that the circular flows took heat away from the keyhole to keep the heat balance. The porosity was observed to form at the tip of the keyhole induced by the impact of the liquid metal flow or at the rear wall of the keyhole induced by the metal vapor.

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